Mini Paper — 1.1 Processors, Input, Output & Storage

Time: ~35 minutes · Total: 30 marks · Answer all questions.

Mark your work against the mark scheme at the end; aim for ~1 minute per mark.

Questions

Q1. State the function of each of the following CPU registers: **(a)** the Memory Address Register (MAR), and **(b)** the Memory Data Register (MDR). **[2] (AO1)**

Q2. Describe the register transfers that take place during the **fetch** phase of the Fetch–Decode–Execute cycle. **[4] (AO1)**

Q3. A CPU has three levels of cache (L1, L2 and L3).

(a) Explain how cache improves the performance of the CPU. **[2] (AO1)**

(b) Describe **one** difference between L1 cache and L3 cache. **[2] (AO1)**

Q4. A team is designing a microcontroller for a washing machine and is deciding between a Von Neumann and a Harvard architecture. Explain **one** advantage of choosing a Harvard architecture for this embedded device. **[3] (AO2)**

Q5. Compare **CISC** and **RISC** processors. In your answer refer to the instruction set and where complexity lies. **[3] (AO2)**

Q6. A research group trains large neural networks that apply the **same calculation across huge data sets**. Explain why a **GPU** is better suited to this work than a general-purpose CPU. **[3] (AO2)**

Q7. Explain what is meant by **virtual memory** and describe **one** drawback of relying on it heavily. **[3] (AO1/AO2)**

Q8. A media company needs secondary storage for two purposes:

- **(a)** the internal drive of laptops used by journalists who travel and film on location, and
- **(b)** long-term, low-cost archiving of very large volumes of old footage that is rarely accessed.

For **each** purpose, recommend a suitable storage technology (magnetic, flash or optical) and justify your choice, evaluating the trade-offs against the requirements. **[8] (AO3)**